PassFrame tuition

Number Codes

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. 64

2. 87

3. 66

4. 89

5. 56

6. 86

7. 324

8. 9

9. 7

10. 345

11. 28

12. 62

13. 8

14. 11

15. 6

Standard

1. 0123

2. 8765

3. 15

4. 5948

5. 43

6. 55

7. 5936

8. 182

9. 83 (F=6, O=15, R=18, E=5, S=19, T=20, so 6+15+18+5+19+20=83)

10. 19

11. 384

12. 54

13. 1802

14. 7890

15. 86 (T=20, U=21, N=14, N=14, E=5, L=12, so 20+21+14+14+5+12=86)

Standard / Hard

1. 337 (2+1=3, 1+2=3, 4+3=7)

2. 55 (reverse 45 to 54, then add 1)

3. 15

4. 22 (4+7=11, then 11×2=22)

5. 93 (4→9, 8→3, wrapping round)

6. 420 (D=4, O=15, G=7, so 4×15×7=420)

7. 18 (reverse 62 to 26, digit sum of 62 is 8, so 26−8=18)

8. 40 (8 is bigger than 5, so multiply: 5×8=40)

9. 654 (the 1st and 3rd digits, 4 and 6, swap places)

10. 26 (6×2=12→12−10=2, 8×2=16→16−10=6, giving 26)

11. 58 (digit sum of 47 is 11, so 47+11=58)

12. 19

13. 65 (shift 45 to 56, then reverse to get 65)

14. 27 (3×9=27)

15. 24 (D=4, O=15, G=7, sum=26, then 26−2=24)

Homework — Answers

Easy / Standard

1. 92

2. 25

3. 45

6. 73

7. 233

8. 10

11. 46

12. 62

13. 11

4. 78

5. 67

9. 8

10. 333

14. 15

15. 7

Standard

1. 9063

2. 5319

3. 18

4. 1593

5. 49

6. 78

7. 4627

8. 368

9. 59 (I=9, S=19, L=12, A=1, N=14, D=4, so 9+19+12+1+14+4=59)

10. 15

11. 261

12. 50

13. 9254

14. 5621

15. 72 (C=3, A=1, N=14, Y=25, O=15, N=14, so 3+1+14+25+15+14=72)

Standard / Advanced

1. 78 (the hidden shift is +3, found from 2→5: 4+3=7, 5+3=8)

2. 21 (the rule is: add the digits, then multiply by 3: 5+2=7, 7×3=21)

3. 26 (the rule is: double each digit: 1×2=2, 3×2=6)

4. 54 (the rule is: shift each digit forward by 1, then reverse the result: 34→45→reversed 54)

5. 28 (the rule: if the second number is bigger, multiply; otherwise subtract — here 7 is bigger than 4, so 4×7=28)

6. 691 (5+1=6, 7+2=9, 8+3=11→wraps to 1)

7. 81 (reverse 27 to 72, digit sum of 27 is 9, so 72+9=81)

8. 46 (3×4=12, then 34+12=46)

9. 574 (the last digit is 4, so the first digit 1 becomes 1+4=5, giving 574)

10. 621 (reverse 326 to 623, the middle digit of 326 is 2, so 623−2=621)

11. 4157 (D=4, O=15, G=7, giving 4-15-7 → 4157)

12. 64 (double 23 to 46, then reverse to get 64)

13. 85 (3+5=8 is the new tens digit, the units digit stays 5, giving 85)

14. 9 ($3^2 = 9$)

15. 17 (double 35 to 70, reverse of 35 is 53, so 70−53=17)